

WEEK 2

Pharma Datasets & Introduction to Claims Analytics

What the manufacturer can actually see — and how to turn codes into a cohort you can defend.

Andi Cupallari · Applied Data Science and AI in Pharma Commercial · Fall 2026 · Monday, September 21

Where we're going tonight

2.5 hours · two live notebooks · one in-class exercise · mentors introduced

0

Welcome back, mentors and teams

Five incentives you found surprising · mentor introductions · team milestones — 15 min

1

The data landscape

Nine categories; medical vs. pharmacy claims vs. EHR vs. syndicated panels; open vs. closed — 20 min

2

Claims 101

Header vs. line · place of service · ICD-10, CPT, HCPCS J-codes, NDC · reading a row — 20 min

3

Rx panel data and projection

Longitudinal vs. cross-sectional · patient capture · how a panel becomes 'national' — 20 min + notebook

4

Building a cohort — live

The ≥ 2 -dx rule, index dates, enrollment; DuckDB; drug events → lines of therapy on 640 DLBCL patients — 40 min

5

RWD quality and re-identification risk

FDA's RWE framework · eight failure modes · HIPAA de-identification · what you may not do — 20 min

6

In-class exercise and next week

Dataset triage · projection by hand · cohort by hand — 25 min

00

Welcome back, mentors and teams

Ten minutes on what surprised you last week, then the people who will supervise your term project.

Five incentives you found surprising — and the data that would prove each one

You were asked to bring one question about a stakeholder incentive. Let's hear five. For each: which dataset from tonight would let you test it?

Typical question from Week 1	The stakeholder mechanism	Tonight's dataset that would test it
"Why would a PBM prefer a higher list price?"	Rebates are a % of WAC; the PBM's revenue rises with list	Plan/formulary intelligence + net-price disclosures (Wk 6)
"Why does a rejected claim matter to the manufacturer?"	Prior auth → abandonment; the fill never happens	Pharmacy claims: REJECTED / REVERSED status, reject codes
"Why is a hospital clinic cheaper for the drug but not the patient?"	340B ceiling price to the entity; patient cost-sharing unchanged	Medical claims: place of service 22 + billing HCO
"How does the manufacturer know its own market share?"	It doesn't — it licenses a projected panel	Rx panel projection (tonight \$3)
"Does a rep visit change prescribing?"	Only if the payer lets the script through	CRM + projected Rx + formulary — Week 3's MMM

Term project: mentors introduced tonight; team rosters confirmed

Mentors are senior practitioners volunteering their time. The team owns scheduling, agendas and follow-ups.



Your first mentor meeting (30–45 min) — an agenda that works

- ▶ Introduce the team and each member's strengths (5 min)
- ▶ Bring two candidate problems from the use-case catalog; ask which is more realistic (15 min)
- ▶ Ask: what data would a real team use for this, and what will we have to synthesize? (10 min)
- ▶ Agree cadence (every two weeks) and a shared channel; send a 5-line recap the same day

Problem statement + data plan (due Oct 5, one page)

- ▶ The decision the analysis informs, and who makes it
- ▶ The cohort or universe: disease codes / HCP specialty / geography — using tonight's template
- ▶ Data: which of the nine categories, synthetic or public, grain, time window
- ▶ First model or analysis you will attempt; what 'good' looks like; risks

Tonight's cohort-definition template (slide 27) is the format your data plan should use.

01

The data landscape

Nine categories cover everything you'll touch. Most projects use three to five. The art is choosing on purpose.

The most consequential thing you do in week one: pick the dataset

Pick wrong and every model downstream inherits the error — and AI now scales that error across thousands of decisions

“In life sciences, data quality has always mattered. In the agentic era, it has become a front-line commercial issue. Poor data quality is no longer simply a reporting inconvenience or remediation task. It can translate into the wrong customer being prioritized, the wrong message being delivered, the wrong channel being selected, or the wrong signal being interpreted as an opportunity. [...] AI can operationalize those errors across hundreds or thousands of decisions simultaneously. The question is no longer whether bad data creates risk. The question is how quickly that risk can scale.”

— IQVIA, “AI Alone Will Not Modernize the Life Sciences Enterprise” (Silvis, 2026)

1 The Week 1 fact

- ▶ The manufacturer has first-party data on none of chooser, payer or consumer
- ▶ Everything it knows about its own market is licensed, sampled and projected

2 The consequence

- ▶ Every dataset is a partial view with a specific blind spot
- ▶ The analyst's job is to know the blind spot before the model does

3 Tonight's skill

- ▶ Name the dataset that answers a question — and the one that can't
- ▶ Build a cohort with a rule you can write down and defend

Nine categories that cover everything you'll touch in pharma commercial

Companion reference in your folder: Pharma_Commercial_Datasets_Reference.docx — one page per category with vendors, grain, latency and gotchas

1 Patient activity

Medical claims · pharmacy claims · EHR · labs ·
specialty pharmacy / hub

2 Prescriber & HCO reference

NPI registry + commercial overlays; HCO / IDN
affiliation graphs

3 Promotion, sales & projection

DDD / wholesaler 867 · projected Rx (Xponent-style) ·
NPA

4 Payer & access

Formulary / UM intelligence (MMIT-style) · plan-linked
claims (APLD)

5 Marketing & engagement

CRM / field activity · digital engagement · consent

6 Genomics & diagnostics

NGS / biomarker testing; precision-medicine sizing

7 Geography, SDOH, mortality

Census / SDOH overlays; death master files

8 Market research & social

Surveys, ad boards, social listening (Wk 7 NLP)

9 Internal finance & ops

Gross-to-net, chargebacks, contracts, inventory

Tonight lives in categories 1 and 3. Weeks 3–5 add 4 and 5; Week 6 adds 7; Week 7 adds 8.

Medical claims, pharmacy claims, EHR, syndicated panel — side by side

Learning objective 1: distinguish them. The last two rows are what interviewers actually ask.

	Medical claims	Pharmacy claims	EHR / EMR	Syndicated Rx panel
Who generates it	Provider bills payer (837P/837I)	Pharmacy adjudicates with PBM (NCPDP)	Clinician documents care	Vendor aggregates pharmacy / clearinghouse feeds
Grain	Service line (visit × procedure × dx)	One row per fill (paid / rejected / reversed)	Encounter, note, lab, order	Pharmacy × NDC × week (or prescriber × product)
What it answers	Diagnoses, procedures, infused drugs (J-codes), site of care, referral paths	Retail/mail/specialty drugs, adherence, access rejections, patient pay	Outcomes, biomarkers, stage, performance status, response	Market share, prescriber ranking, trend — projected to national
Blind spot	Inpatient drugs bundled in DRG; no outcomes; open-claims gaps	Nothing infused in clinic; cash fills; no diagnosis on the row	Only the systems in the network; free text; not projectable	Sample, not census; no patient link (unless longitudinal); restated
Typical vendors	IQVIA, Komodo, Symphony, HealthVerity, Optum (closed)	IQVIA LRx, Symphony, PBM-direct	Flatiron, ConcertAI, Tempus, Epic Cosmos	IQVIA Xponent/NPA, Symphony PHAST

Open vs. closed claims — pick on purpose, not by default

The single most common source of a confidently wrong share number: numerator from one, denominator from the other

Open claims (clearinghouse / switch)

- ▶ **Source:** the pipes between providers, pharmacies and payers; whichever claims pass through a partner switch
- ▶ **Strength:** breadth — very large patient counts, fast (days), all payers
- ▶ **Weakness:** you never know what you're missing: a patient's claims from a non-partner pipe are simply absent
- ▶ **Use for:** patient finding, HCP targeting, market sizing with projection

Closed claims (payer / enrollment-based)

- ▶ **Source:** a payer's or PBM's own adjudication file, with enrollment (eligibility) records
- ▶ **Strength:** completeness — you know exactly when the patient was covered, so absence of a claim means absence of care
- ▶ **Weakness:** narrower (one payer's members), slower (quarterly), skewed by that payer's mix
- ▶ **Use for:** outcomes, persistence, cost, anything with a denominator

Tonight's DLBCL file ships with an enrollment table — that is what makes it closed-claims-like. Watch what it does to the cohort (slide 30).

Rule of thumb: when the question has the word “rate” in it, ask which denominator you have.

Five-question triage — run any analysis through this before licensing anything

Reusable for the term project data plan

1

Decision

What decision does this inform — brand strategy, targeting, access, medical, forecasting? Who acts on it?

2

Population

Sizing a universe (needs projected / open) or measuring within a defined cohort (needs observed / closed)?

3

Granularity

Patient-level (claims, EHR) • HCP-level (projected Rx, CRM) • HCO-level (DDD) • geo-level (census)?

4

Time

Real-time decisioning (CRM, digital) • weekly brand tracking (panels) • quarterly outcomes (closed claims)?

5

Cost & rights

License cost, refresh, what you may link, what you may not re-identify, and whether the vendor allows model training

02

Claims 101

Header vs. line, place of service, and the four code systems that join every dataset in this course.

Anatomy of a medical claim: one header, many lines

The 837 professional (office) and institutional (facility) formats differ, but the shape is the same

HEADER — one per claim

- ▶ **claim_id** · patient token
- ▶ **billing provider** / HCO (NPI, tax ID)
- ▶ **payer**, plan, member ID
- ▶ **dates of service** (from / to)
- ▶ **place of service** (11 office · 22 hospital outpatient · 21 inpatient)
- ▶ up to 12 diagnosis codes (ICD-10-CM); dx1 = primary
- ▶ **total billed** · allowed · paid · patient responsibility

LINES — one per service

- ▶ **procedure code**: CPT (99214 office visit, 96413 chemo infusion) or HCPCS Level II (J9312 rituximab, Q2041 axi-cel)
- ▶ **units**: J-code units are dose units (J9312 = 10 mg); 20 units = 200 mg
- ▶ **rendering provider NPI** (the clinician), which may differ from the billing NPI
- ▶ **dx pointer**: which of the header diagnoses this line is for
- ▶ **line allowed / paid amounts; revenue code** on facility claims (0636 = drugs requiring detail coding)
- ▶ **modifiers**: JW (drug wastage), 25, 59 ...

In vendor extracts the header is usually denormalized onto every line — which is why the same diagnosis appears 31,000 times in tonight's file.

The code systems — the keys that join every dataset in this course

Glossary entries exist for all of these (Week 1 handout). Tonight: what each one looks like and what it is for.

Code	Looks like	Identifies	Owner	Where it appears	Watch out for
ICD-10-CM	C83.30 • E11.65	Diagnosis (why)	CDC / CMS	Medical claim header	Rule-outs coded as dx; specificity varies by coder; annual updates (Oct 1)
CPT	99214 • 96413	Procedure / service (what)	AMA	Medical claim line	E/M levels reflect billing, not severity
HCPCS Level II	J9312 • Q2041 • G0463	Drugs, biologics, supplies	CMS	Medical claim line	New drugs sit in J3490 / C9399 'unclassified' for ~6 months
NDC	0002-1434-80 → 00002143480	Product × labeler × package	FDA (labeler assigns)	Pharmacy claim; some medical lines	10-digit vs. 11-digit formats; NDC recycled; biosimilars have their own
NPI	10 digits, Luhn check	Individual (type 1) or organization (type 2)	CMS NPPES	Rendering, billing, prescriber	Same person, several organizations; locums; 'missing' = billed under group
Place of service	11 • 22 • 21 • 19	Site of care	CMS	Medical claim header	22 vs 19 (off-campus outpatient) changes payment and 340B
DRG	MS-DRG 840	Inpatient stay bundle	CMS	Institutional claims	Drugs inside a DRG have no line — invisible
Plan / payer IDs	vendor-specific	Who paid	Vendor	Both claim types	Payer mapping to channel (Commercial / Medicare / Medicaid) is a crosswalk, not a fact

Anatomy of a pharmacy claim: one row per adjudication event

NCPDP D.0 — the pharmacy sends it, the PBM answers in seconds. Week 1's GLUCOREN tour was this table.

Field	Example	Why an analyst cares
patient token • DOB year • sex • ZIP3	P00017 • 1968 • F • 100	Longitudinal link across fills; de-identified (HIPAA Safe Harbor)
fill_date • days_supply • quantity	2026-03-04 • 30 • 30	Refills, gaps, persistence, 90-day normalization (mail order)
NDC (11-digit)	00054475625	Product, strength, package → molecule → market
prescriber NPI	1234567893	Joins to CRM targets and to specialty; the 'chooser'
pharmacy NPI / NCPDP ID • type	PH3911 • Specialty	Channel; PBM-owned mail/specialty steering
plan / BIN-PCN • payer type	PL-COM-04 • Commercial	The access gate; channel mix
claim status • reject code	PAID • REJECTED (75) • REVERSED	Prior auth, step edit, abandonment
ingredient cost • dispensing fee • patient pay • plan pay • copay-card	\$410 • \$2 • \$45 • \$367 • \$30	Who paid what at the counter — not the manufacturer's net
DAW code • refill number	1 • 3	Brand-vs-generic substitution; new vs. refill (NRx / TRx)

There is no diagnosis on a pharmacy claim. To know why the drug was prescribed, you link to medical claims through the patient token — which is why longitudinal linkage matters so much.

Which benefit paid for the drug decides which file you'll find it in

The question that separates a primary-care analyst from an oncology analyst

Medical benefit — “buy and bill”

- ▶ Physician / hospital buys the drug (wholesaler, or specialty distributor), administers it, bills the payer
- ▶ Appears on the medical claim as a HCPCS J/Q-code + units + administration CPT (96413...)
- ▶ Infused mAbs, chemotherapy, CAR-T, many biologics; site of care = office (11) or hospital outpatient (22)
- ▶ Payment = ASP + 6% (Medicare); 340B in eligible hospitals; drug price invisible on the row
- ▶ Inpatient: usually bundled in the DRG → no line at all

Pharmacy benefit — dispensed

- ▶ Pharmacy dispenses, adjudicates with the PBM, is reimbursed; patient pays copay/coinsurance
- ▶ Appears on the pharmacy claim as an NDC + days' supply + quantity
- ▶ Oral agents, self-injectables, specialty-pharmacy drugs (including some oncology orals like lenalidomide, selinexor)
- ▶ Formulary tier, prior auth, step edits, copay cards all live here
- ▶ Rejections and reversals are visible — the access story is in this file

DLBCL tonight: rituximab, CHOP, polatuzumab, bispecifics, CAR-T → medical (J-codes). Prednisone, lenalidomide, selinexor → pharmacy (NDCs).
You cannot see a regimen until you stack both files by patient and date.

Reading two rows from tonight's file like an analyst

From dlbc_medical_claims.csv and dlbc_pharmacy_claims.csv (synthetic)

claim_id	patient	service_date	provider_npi	pos	claim_type	code_type	code	code_desc	diagnosis	units	allowed	payer
M100000005	PT00000	2025-06-26	1874414982	22	drug_admin	HCPCS	J9312	rituximab	C83.30	20	17714.88	Medicare
M100000005R	PT00000	2025-06-26	1874414982	22	drug_admin	HCPCS	J9312	rituximab	C83.30	20	17714.88	Medicare

- ▶ **pos 22 + J9312 + 20 units:** 200 mg of rituximab infused in a hospital outpatient department — medical benefit, possibly 340B
- ▶ **C83.30 on a drug line:** the dx is copied from the header; it does not mean the drug 'treated' DLBCL that day
- ▶ **the R suffix:** a resubmission — same service, same date, same amount. Count it once or you double the dose
- ▶ **\$17,714.88 allowed:** what the payer agreed to pay the provider; says nothing about the manufacturer's net price

rx_id	patient	fill_date	ndc	drug
P700000001	PT00000	2025-06-26	00054475625	prednisone

days 14 · qty 14 · prescriber 1874414982 · PH3911 · Medicare

- ▶ **Same patient, same day, different file:** the oral steroid that completes R-CHOP
- ▶ **No diagnosis here** — the link to C83.30 is through the patient token and the date
- ▶ **days_supply 14:** the prednisone component of a 21-day cycle

03

Rx panel data and projection

How a sample of pharmacies becomes “national TRx” — and the two capture problems hiding inside the number.

Longitudinal vs. cross-sectional — same fills, different questions

The first thing to ask of any Rx dataset: is there a persistent patient token?

Cross-sectional (aggregate)

- ▶ **Grain:** pharmacy × product × week, or prescriber × product × week
- ▶ **No patient identity** — counts only
- ▶ **Answers:** how many scripts, which prescribers, market share, trend; NRx vs. TRx (refill flag)
- ▶ **Cannot answer:** new-to-brand vs. switch, persistence, concomitant therapy, line of therapy
- ▶ **Projectable to national** with stratified factors — this is the weekly brand report

Longitudinal (patient-level)

- ▶ **Grain:** de-identified patient × fill (and × medical claim if linked)
- ▶ **Persistent token** across pharmacies, payers and years
- ▶ **Answers:** NBRx, source of business, adherence, persistence, patient journey, cohorts, LOT
- ▶ **Capture problem:** the patient is only seen where the panel sees them; a fill at a non-panel pharmacy looks like a gap
- ▶ **Usually not projected;** used for rates within the observed population

NRx ≠ NBRx. NRx = no fill of this product at this pharmacy in ~12 months (cross-sectional). NBRx = the patient has never been on this brand anywhere (needs a token). Interviewers ask this. Tonight's projection tour shows why.

Two coverage problems: pharmacy coverage and patient capture

Both are sampling problems; they bias different metrics

Pharmacy coverage — share of dispensing volume the panel sees

- ▶ **Real U.S. retail panels capture ~90%+ of retail volume** (Week 1 glossary); tonight's synthetic panel is deliberately thinner (~60%) so that projection visibly matters
- ▶ **Uneven by stratum:** chains join panels more than independents; mail order is 2-3 facilities; specialty pharmacies are few and closed
- ▶ **Biases:** levels (fixable by projection) and, when a large member joins or leaves, trends (not fixable without a restatement)

Patient capture — share of a patient's care the data sees

- ▶ **A patient filling at panel and non-panel pharmacies** looks non-adherent; a patient whose oncologist is at a non-partner IDN looks untreated
- ▶ **Biases rates, not levels:** persistence, switching, time-to-treatment, share of patients on 2L — all understated or distorted
- ▶ **Mitigations:** closed-claims sources with enrollment; require ≥ 1 claim per quarter as an 'activity' criterion; report capture rate alongside every rate

Projection is a stratified ratio estimator — nothing more mysterious than that

Numbers from tonight's synthetic GLUCOREN panel: 233 panel pharmacies out of 388, six ZIP3s, retail / specialty / mail

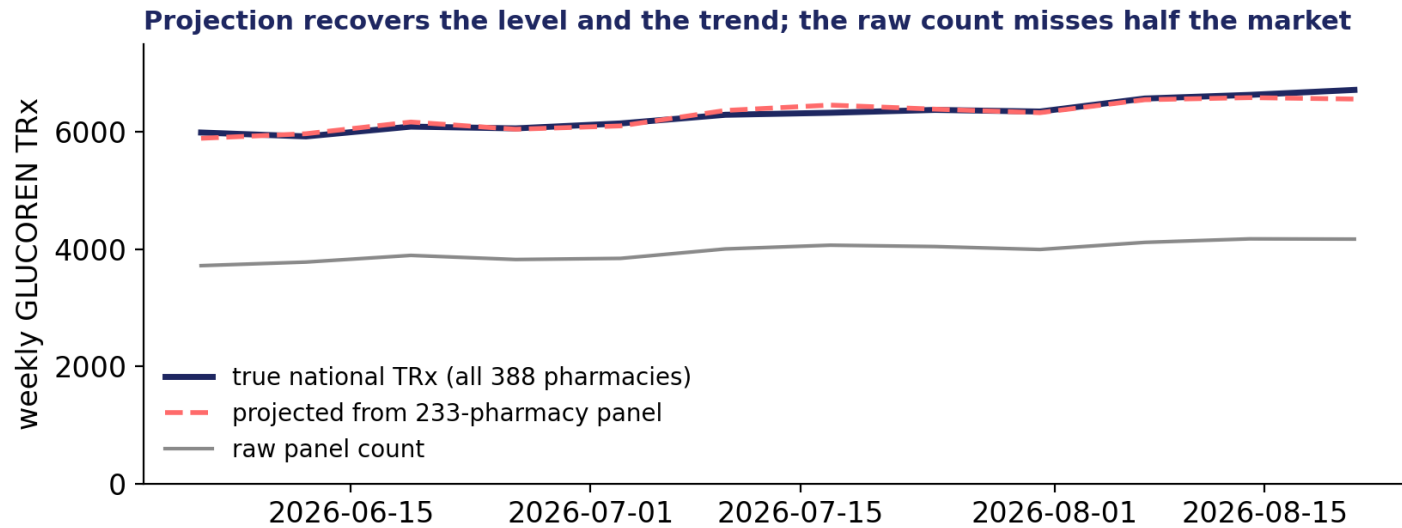
Stratum (ZIP3 × type)	Panel pharmacies	Universe pharmacies	Coverage (volume)	Factor = 1 / coverage
100 • Retail	45	70	68%	1.47
070 • Retail	29	51	62%	1.60
070 • Specialty	1	4	28%	3.63
104 • Specialty	4	4	100%	1.00
117 • Specialty	0	4	0%	— (borrowed)
MAIL • Mail	1	3	31%	3.25

- ▶ **Strata** are chosen so that brand share is homogeneous inside them: geography × outlet type × size
- ▶ **Factors** come from a census of volume (wholesaler / 867 data), not from the panel
- ▶ **Uncovered cells** are borrowed from neighbours — a judgment call that never appears on the report
- ▶ **Single-member strata** (one mail-order facility, factor 3.25) are the fragile part
- ▶ **Small cells** (one ZIP3 × one week × one specialty pharmacy) are where HCP-level targeting lives — and where the noise is

```
-- projected brand TRx, per stratum and week
projected = panel_brand_trx × (universe_volume / panel_volume)
national  = Σ strata    -- then impute the uncovered ones
```

On synthetic data we can grade the projection. In real life you reconcile instead.

Instructor truth: all 388 pharmacies. Students see only the 233-pharmacy panel.



- ▶ **Level:** within ~1–2% each week on this (well-behaved) synthetic market
- ▶ **Trend:** preserved — projection multiplies, it doesn't smooth
- ▶ **Raw panel count:** ~60% of the truth; anyone reading it as 'sales' is off by half
- ▶ **Where it breaks in practice:** panel churn, a chain changing its data feed, new-outlet types (e.g., a manufacturer-direct channel), cash-pay and 340B contract pharmacy flows
- ▶ **Reconcile:** projected TRx × units vs. wholesaler shipments (DDD/867) vs. your own net sales — three numbers that must tell one story

The syndicated reports a brand team reads every Monday

What each one measures, and what it silently assumes

Report family	Measures	Source	Grain	Trust it for	Don't trust it for
DDD / 867 sales-out	Units shipped from wholesalers to outlets (pharmacies, clinics, hospitals)	Wholesaler EDI 867 / 852	Outlet × NDC × week	Demand at the outlet level; territory sales credit; reconciling to revenue	Patient demand (inventory swings, returns); who prescribed
Projected Rx (Xponent-style)	Prescriptions by prescriber and product, projected to national	Retail / mail / LTC pharmacy panel	Prescriber × product × week	Targeting, share, NRx / TRx trend, decile ranking	Small prescribers (imputation), specialty-heavy brands, buy-and-bill drugs (not there at all)
NPA / national audit	National Rx volume and value by product / market	Same panel, national projection	Product × market × month	Market size, share, price-volume	Sub-national or sub-segment cuts
Patient-level LRx / APLD	Longitudinal fills and (if linked) medical claims	Pharmacy / clearinghouse + tokenization	Patient × event	Source of business, persistence, journeys, LOT	National levels (capture), rare events in small cells
Specialty pharmacy / hub	Referral → enrollment → fill for specialty brands	Manufacturer's own SP network / hub	Patient × status × date	Time-to-fill, PA outcomes, abandonment for your brand	Competitors; anything outside your network

When to trust a syndicated number — five questions to ask the vendor (and yourself)

Learning objective: “syndicated reports and when to trust them”

1

Coverage of the strata I care about?

National retail: fine. One specialty pharmacy in one metro: no. Ask for coverage by outlet type and geography.

2

What changed in the panel this period?

A large member joining or leaving moves 'national' numbers overnight. Ask for the panel-change log and restatements.

3

Cross-sectional or patient-level?

TRx trend, share, targeting: cross-sectional is fine. Persistence, switching, NBRx, LOT: patient-level only.

4

What does it reconcile to?

Projected TRx × units ↔ DDD shipments ↔ your net sales, with lags. If they disagree, one of them is wrong — find out which.

5

What are the lag and restatement rules?

Last week's number will be restated as late claims arrive. Compare mature weeks to mature weeks; never the freshest week to a prior one.

Notebook: [Week_02_Rx_Panel_Projection_Tour.ipynb](#) — 10 minutes, run it now.

04

Building a cohort — live

From ICD-10 codes to 482 defensible DLBCL patients, then to lines of therapy. Notebook:
`Week_02_DLBCL_Cohort_LOT_Notebook.ipynb`

Why DLBCL tonight — an oncology market where the data problem is the whole problem

Diffuse large B-cell lymphoma: the most common aggressive non-Hodgkin lymphoma; ICD-10-CM C83.3x

30–40%

of non-Hodgkin lymphoma cases are DLBCL; incidence rises with age (IQVIA)

2 of 3

eligible relapsed / refractory LBCL patients are never referred to a CAR-T treatment center (IQVIA survey research)

< 2 in 10

ultimately receive CAR-T — the gap between clinical potential and the real world

~1 in 4

referred patients drop out at the treatment center after screening (ineligible by then)

- ▶ **Why it is a claims problem:** almost every drug is infused (J-codes), care moves between community oncologists and academic centers (two NPIs, two sites of care), and the commercial question — who reaches 2L and 3L, and where — is a patient-journey question.
- ▶ **What the brand team needs:** 1L / 2L / 3L+ regimen shares, time to next treatment, the referral funnel by site of care, and the payer mix at each line.
- ▶ **What IQVIA did to build its referral study:** “IQVIA leveraged 2023 claims data to define high and low thresholds for referring oncologists and treaters in addition to triangulating referral and treatment volumes.” Claims are the sampling frame for everything else.
- ▶ **Ethics reminder:** these are real diseases and real drugs; the patients tonight are synthetic. The methods are the ones used on real data under the governance rules in §5.

A cohort is a rule, not a fact — write the rule down before you write the query

Use this table in every project (and in your Oct 5 data plan)

Element	Tonight's choice	The trade-off you are making
Disease code(s)	ICD-10-CM C83.30 (real projects: the full C83.3* family, ± histology from EHR)	Narrow codes miss patients; broad codes admit neighbours (follicular C82.x)
Confirmation rule	≥ 2 claims with the code on different days, the first two ≤ 365 days apart	One claim may be a rule-out; two a year apart may be surveillance of an old disease
Index date	First qualifying diagnosis day	Everything (LOT, time-to-event) is measured from here; a vendor's 'dx date' is not yours
Baseline (washout)	≥ 180 days continuous enrollment before index, no gap > 45 days	You lose newly insured patients; you gain the ability to see prior treatment
Follow-up	Enrolled ≥ 1 day after index; censor at coverage end	Patients who leave the panel are censored, not dead
Activity / capture	(optional) ≥ 1 claim of any kind per 6 months	Guards against patients the data only half-sees
Exclusions	(optional) prior lymphoma code before index; age < 18	Incident vs. prevalent cohort — say which one you built

The ≥ 2 -dx rule — and the trap inside it

Sensitivity (catch every real patient) vs. specificity (admit no impostor). You choose the balance for the purpose.

1 Why two claims

- ▶ A single diagnosis code is often a work-up: 'rule out lymphoma' gets coded as lymphoma
- ▶ In tonight's file: 70 patients have exactly one C83.30 claim; 70% of them have a later benign code (R59.0)
- ▶ Requiring two claims on different days removes all of them

2 Why a window

- ▶ Two claims years apart can be surveillance of a cured disease, a re-coded history, or a data error
- ▶ 25 decoys have two claims 400–800 days apart
- ▶ Requiring the first two within 365 days removes them

3 The trap

- ▶ “First two within 365 days” is brittle: one stray back-dated claim pushes a real patient's first two dx days years apart
- ▶ 11 real patients are lost that way tonight
- ▶ Robust variant: any two dx days within 365 days — try it

Market sizing tolerates false positives (use a loose rule). An outcomes study or a payer dossier does not (tight rule, EHR confirmation). Same data, different cohorts.

The rule in DuckDB — window functions do the work

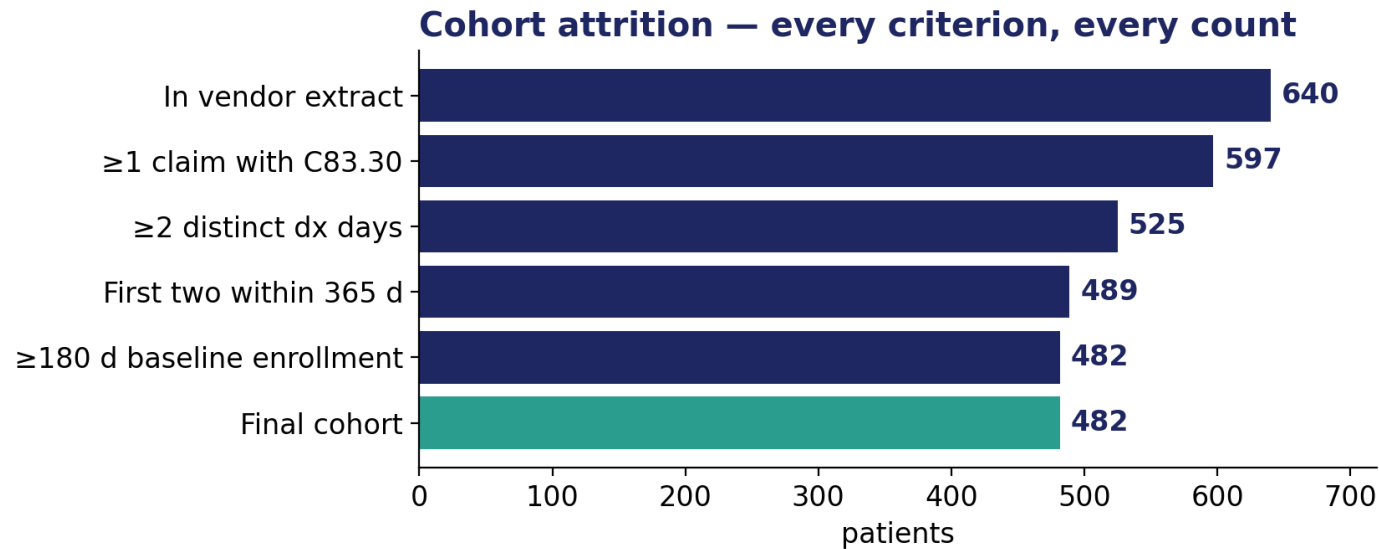
DuckDB runs SQL on DataFrames or CSVs with no server; the same query runs on Snowflake / Databricks at work

```
WITH dx_days AS (
    -- distinct diagnosis days
    SELECT DISTINCT patient_id, service_day
    FROM clean WHERE diagnosis_code = 'C83.30'),
ranked AS (
    SELECT patient_id, service_day,
           ROW_NUMBER() OVER (PARTITION BY patient_id
                               ORDER BY service_day) AS rn
    FROM dx_days),
first_two AS (
    SELECT patient_id,
           MAX(CASE WHEN rn=1 THEN service_day END) AS dx1,
           MAX(CASE WHEN rn=2 THEN service_day END) AS dx2,
           COUNT(*) AS n_dx_days
    FROM ranked GROUP BY patient_id)
SELECT patient_id, dx1 AS index_date, n_dx_days,
       dx2 IS NOT NULL AND date_diff('day', dx1, dx2) <= 365 AS confirmed
FROM first_two;
```

- ▶ **clean** is the medical table after de-duplication and removal of adjustment lines — never run the rule on raw
- ▶ **DISTINCT days**, not claims: three lines on one visit are one diagnosis day
- ▶ **ROW_NUMBER** gives the ordering; MAX(CASE ...) pivots the first two into columns
- ▶ **The pandas version** in the notebook produces the same 489 patients — we assert it
- ▶ **Then join enrollment** for baseline and follow-up; that is Python in the notebook, SQL at work

Every cohort ships with an attrition table — no funnel, no trust in the denominator

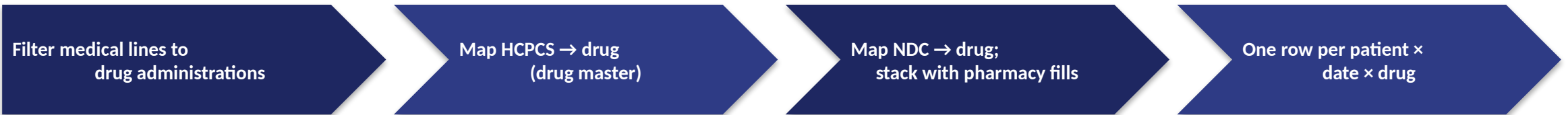
Tonight's numbers (synthetic file, 640 patients)



- ▶ **640 → 597**: 43 patients never had a C83.30 claim (follicular lymphoma C82.x — a vendor threw in the neighbourhood)
- ▶ **597 → 525**: 72 single-claim patients — the rule-outs
- ▶ **525 → 489**: 25 'far apart' decoys, plus 11 real patients lost to a back-dated claim
- ▶ **489 → 482**: the enrollment criterion — patients we could not observe for 180 days before index
- ▶ **Grade (because we can)**: sensitivity 96%, specificity 99%, PPV 99.8%. In real life you validate a sample against charts instead

From codes to drug events: stack J-codes and NDCs into one timeline per patient

Four steps, one crosswalk, one sanity check



- ▶ **Crosswalk:** dlbc1_drug_master.csv — 23 codes → drug → class. At work: Medi-Span / FDB / RedBook plus a curated J-code list
- ▶ **Exclusions:** lymphodepleting cyclophosphamide before CAR-T is conditioning, not a line
- ▶ **Unmapped codes** must be zero or explained: J3490 'unclassified' hides new launches for months
- ▶ **Sanity check:** rituximab and cyclophosphamide must top the count — they are in nearly every 1L regimen

```
drug events in cohort, on/after index: 15,367
medical 12,430 · pharmacy 2,937
```

rituximab	3048	
cyclophosphamide	2860	
prednisone	2859	-- pharmacy
doxorubicin	2647	
vincristine	2093	
polatuzumab vedotin	878	
epcoritamab	232	
glofitamab	180	

Line of therapy: there is no canonical definition — so document yours, every time

Five rules, three parameters. Change W or G and the shares change.

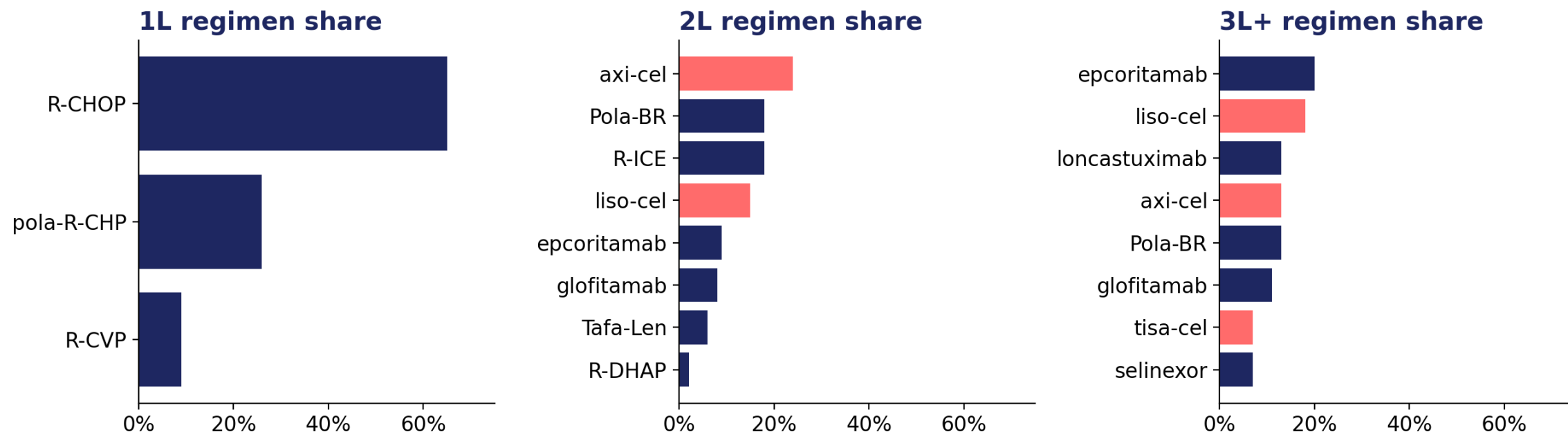
#	Rule	Parameter	Why
1	First anti-lymphoma drug on/after index starts LOT 1	—	Index anchors everything
2	Drugs first given within W days of a LOT start belong to that LOT's regimen	W = 28 d	R-CHOP components arrive over a cycle; too short a window splits a regimen
3	A gap of more than G days with no anti-lymphoma drug ends the LOT; the next drug starts a new one	G = 90 d	Distinguishes a treatment break from progression; too short creates phantom lines
4	A CAR-T infusion always starts a new LOT	—	Even inside a gap window — it is a different intent
5	Steroids alone never start a LOT	7-day look-back	Prednisone rides along with the chemo it accompanies

► **Regimen naming:** the drug bag within W days is matched to a dictionary (R-CHOP, pola-R-CHP, R-ICE, Pola-BR, bispecifics, CAR-T...) by Jaccard similarity. R-mini-CHOP is invisible — dose is not in claims.

► **Maintenance, ASCT and radiation** each need a rule too (tonight: ignored). Consensus across IQVIA / ZS / Adelphi methodologies, but none is 'right'.

What the analysis recovers: the 1L / 2L / 3L+ landscape

482 patients • 676 lines • CAR-T in coral



- ▶ **1L:** R-CHOP 65% with pola-R-CHP the growing #2 (post-POLARIX). 2L fragments: CAR-T (axi-cel, liso-cel) vs. salvage chemo ahead of transplant vs. Pola-BR. 3L+: bispecifics lead.
- ▶ **Next cuts the brand team asks for:** age band (CAR-T eligibility falls with age: 20% under 60 → 0% over 80 tonight), payer, site of care (POS 21/22 vs 11), academic vs. community NPI.

TTNT — the closest a claims file gets to an outcome, and the easiest one to get wrong

TTNT for line n = start of line n+1 – start of line n

Line	Patients with a next line	Mean (d)	P25	Median	P75
1L → 2L	136	450	351	438	558
2L → 3L	56	283	203	272	368

Why brand teams care

- ▶ Shorter TTNT after a competitor's regimen = a commercial argument (and a medical one)
- ▶ It is a proxy for progression-free survival — with claims-grade noise
- ▶ Payers read it too; it appears in dossiers

Censoring is not death

- ▶ The table above counts only patients who *had* a next line
- ▶ Missing: patients cured, patients who died, and patients who lost coverage (37 tonight) — three very different stories, one absence
- ▶ Proper method: Kaplan-Meier with censoring at coverage end (Week 6); pair with mortality data
- ▶ Never present a mean TTNT without saying who is excluded
- ▶ Same for 'share of patients reaching 2L': the denominator must be patients with enough follow-up to reach it

05

RWD quality and re-identification risk

What regulators mean by real-world evidence, eight ways data quietly lies, and what you may never do with a patient token.

RWD vs. RWE — FDA's framework is the vocabulary everyone uses

Reading: FDA, “Framework for FDA's Real-World Evidence Program” (Dec 2018), exec summary + §II; optional: 2024 final guidance on RWD/RWE

Definitions (FDA framework)

- ▶ **RWD:** “data relating to patient health status and/or the delivery of health care routinely collected from a variety of sources” — claims, EHR, registries, devices, apps
- ▶ **RWE:** “clinical evidence about the usage and potential benefits or risks of a medical product derived from analysis of RWD”

The three questions FDA asks of any RWE

- ▶ Are the RWD fit for use? (relevance + reliability: accuracy, completeness, provenance)
- ▶ Can the study design provide adequate scientific evidence to answer the question?
- ▶ Does the study conduct meet FDA regulatory requirements (e.g., data integrity, monitoring)?

“Sponsors face a recurring tension: the most accessible datasets are rarely the most fit-for-purpose, and the most fit-for-purpose datasets often require investment in curation, linkage, or chart abstraction long before the SAT reads out. [...] Regulators describe ‘fit-for-purpose’ data that have sufficient quality and relevance for the intended regulatory use. This ensures the data is suitable for regulatory decisions without necessarily requiring data to be close to perfect as it is expected for RCT data.”

— IQVIA, “Bridging the Evidence Gap: External Comparator Studies from Design to Reimbursement” (2025)

Eight ways claims data quietly lies to you — all eight are in tonight's file

Learning objective 3: identify common data-quality risks in RWD. Run these checks before every analysis.

Failure mode	What it looks like	What it does to your answer	The check
Duplicate lines	Resubmissions with identical content (claim_id + 'R')	Doubles doses, visits, cost	Duplicate on (patient, date, NPI, code, dx, amount)
Adjustments / reversals	Negative or zero allowed amounts	Distorts cost; phantom events	Filter amount ≤ 0 ; keep a log
Missing keys	Blank rendering NPI (~1.5%)	Prescriber attribution breaks; targets lost	Null rate per key; fall back to billing NPI
Neighbour codes	C82.x, R59.0 in a 'DLBCL' pull	Cohort inflation	Code frequency table before anything else
Rule-out coding	Single dx claim then a benign code	False positives	≥ 2 -dx rule; look at what follows
Back-dated / mis-dated	A visit 900 days before every other claim	Index dates wrong; patients lost	Claims before vendor dx date; outlier gaps
Coverage gaps / loss	Enrollment ends after dx (37 patients)	Censoring read as death or cure	Enrollment join; report capture
Lag tail	Last 2–3 months thin	'The brand is declining!'	Compare mature periods; completeness factors

Fit-for-purpose: the same question, five data sources, five answers

IQVIA's 'data continuum' for external-comparator studies — the logic applies to any RWD choice

Data source	Relevance	Clinical depth	Time to source	Cost	Commercial analog
Literature	Limited ability to match population	Low-Medium	Short	Low	Analyst reports, published shares
Administrative databases (claims)	Limited to the scope of the billing task; likely lacks clinical depth	Low	Medium	Medium	Tonight — breadth, no outcomes
Patient registries	Suitable cohorts and clinical data; can't alter capture	Medium-High	Medium	Medium	Disease-society registries, hub data
EHR	'Research-ready' digital records; good clinical detail	Medium-High	Medium-Long	Medium	Flatiron-style oncology EHR
Chart review	Gold standard for richness; slow and expensive	High	Long	High	Manual validation of a claims cohort

"Fit-for-purpose data first. The comparator cohort is only as strong as the data that defines it. Prioritise sources with clinical depth, longitudinal follow-up, and documented data quality over sources that are merely accessible. Chart review remains the gold standard for data richness; registry data offers scale." — IQVIA, Bridging the Evidence Gap

HIPAA de-identification: why your patient is a token, an age band and a ZIP3

Two legal routes under the HIPAA Privacy Rule (45 CFR §164.514). Vendors use the second.

Safe Harbor — remove 18 identifiers

- ▶ Names • geographic units smaller than a state (ZIP3 allowed if population > 20,000)
- all dates except year (and ages ≥ 90 → '90+') • phone, fax, email • SSN • medical record, plan, account numbers • certificate / license numbers • vehicle and device IDs
- URLs, IPs • biometrics • full-face photos • any other unique identifier
- ▶ Mechanical and safe — but it destroys dates, and claims analytics runs on dates

Expert Determination — a statistician certifies 'very small' risk

- ▶ Keeps exact dates, ZIP3, age, prescriber — with controls: tokenized IDs, small-cell suppression, contractual bans on re-identification and linkage
- ▶ This is what commercial claims/LRx datasets are: a certified risk, not zero risk
- ▶ Tokenization: a one-way hash of name + DOB + sex (± other fields) so two vendors' files can be linked on the same patient without either knowing who they are (deterministic vs. probabilistic linkage)

“Pseudonymization replaces sensitive identifiers with unique, random pseudonyms that have no intrinsic meaning [...] Depending on the design, pseudonymization can support deterministic linkage (the same input always maps to the same output) or probabilistic linkage.” — IQVIA, Trustworthy AI/ML for Patient Analytics

Re-identification risk: quasi-identifiers, small cells, and what AI adds

Learning objective 3, second half

1 Quasi-identifiers

- ▶ Individually harmless fields that combine into a fingerprint: ZIP + birth date + sex uniquely identified ~87% of the U.S. population in Sweeney's classic 2000 study
- ▶ In claims: rare diagnosis × ZIP3 × exact dates × prescriber is often unique
- ▶ Tonight's file: one CAR-T patient in one state on one date — that is a small cell

2 Linkage attacks

- ▶ Join a de-identified file to a public one (voter rolls, obituaries, social posts, a physician's practice website)
- ▶ The famous cases: Massachusetts governor (1997), Netflix Prize (2007), NYC taxi medallions (2014)
- ▶ Every added dataset lowers k — the number of people who share a record's profile

3 What AI adds

- ▶ Models can infer sensitive attributes and reconstruct inputs from outputs; inference and correlation attacks work on 'de-identified' features
- ▶ Defenses: aggregation, dimensionality reduction, noise / differential privacy, federated training, monitoring
- ▶ Emerging standards: ISO/IEC 42001, NIST AI RMF

"Even relatively sparse information can yield sensitive inferences. [...] Adversarial actors can exploit poorly abstracted model inputs or outputs through inference, reconstruction, or correlation attacks." — IQVIA, Trustworthy AI/ML for Patient Analytics and Research

What you may — and may never — do with patient-level data

These are contract terms in every data license and policy in every company. They will be on the midterm.

Always

- ▶ Suppress small cells (< 11 patients is the common floor; some contracts use 5 or 20) in anything shared
- ▶ Keep the cleaning log and the cohort definition with the result
- ▶ Report capture / coverage next to every rate
- ▶ Use synthetic data for development and demos (as tonight)

Only with governance

- ▶ Linking two patient-level sources (needs a tokenization vendor and a determination)
- ▶ Prescriber-level reporting (state laws: e.g., Vermont's restrictions; AMA PDRP opt-outs)
- ▶ Training ML models on licensed data (check the license's model-training clause)
- ▶ Sharing extracts with a partner or agency

Never

- ▶ Attempt to re-identify a patient — even 'to check'
- ▶ Paste patient-level rows into an external AI tool (course AI policy, Week 1)
- ▶ Use patient-level data to target a patient directly (marketing to individuals from claims)
- ▶ Contact a prescriber about a specific identified patient
- ▶ Keep a copy after the license ends

Term projects: use the synthetic course datasets or public data (CMS Part D prescriber files, NPPES, FDA NDC directory, SEER). No licensed or employer data.

Exercise: triage, project, cohort (25 min, groups of 3–4)

Handout: Week_02_InClass_Exercise. One group presents each part in the debrief.

A Part A • Dataset triage (8 min)

- ▶ Six real questions a GLUCOREN brand team asked this year
- ▶ For each: which dataset(s) from the taxonomy, which grain, open or closed, cross-sectional or longitudinal — and the blind spot you'd flag

B Part B • Projection by hand (8 min)

- ▶ Three strata, panel and universe volumes, panel brand TRx → factors → projected TRx
- ▶ Then: the largest panel chain in one stratum leaves the panel. Recompute. What changed, and what would the brand report say happened?

C Part C • Cohort by hand (9 min)

- ▶ Ten patients' diagnosis claims and coverage spells on one page
- ▶ Apply tonight's template: who qualifies, index date, who is censored
- ▶ Then: which combination of columns in the de-identified extract would re-identify patient 7?

Answer key in the instructor package; students receive the worked solution after the debrief.

Before Week 3 (Sep 28): readings, mentor meeting, environment

Week 3: Marketing & Media Mix Modeling and ROI optimization — adstock, saturation, base/incremental. Homework 1 released (due Oct 12).

Do

- ▶ **Read:** FDA, “Framework for FDA's Real-World Evidence Program” — executive summary and §II ([fda.gov/media/120060/download](https://www.fda.gov/media/120060/download)).
- ▶ **Read:** Week 2 reading pack (curated white-paper excerpts, ~35 min) and the Claims Data Field Guide (reference — skim, keep).
- ▶ **Watch (optional, PMSA):** “Improving Completeness and Accuracy of Real World Data”; “Combine Advanced Analytics and RWD to Identify Treatment Drivers.”
- ▶ **Meet your mentor** by Sep 28; send the 5-line recap. Problem statement + data plan due Oct 5.
- ▶ **Run both notebooks** on your own machine; try the two 'not graded' variations at the end of each.

Bring

- ▶ **Your cohort definition** for the term project, in the slide-27 template — even if the disease/HCP universe is a guess.
- ▶ **One number** from a syndicated-style report you have seen (news, an earnings call, a vendor site) and the question you would ask about how it was projected.
- ▶ **For Week 3:** install PyMC or statsmodels; Homework 1 uses a synthetic MMM dataset with bursty conference spend.

Office hours Mondays 5:00–6:45 PM • ac5562@columbia.edu

Questions?

Five things to leave with:

- 1 · Pick the dataset on purpose (decision · population · grain · time · rights).
- 2 · Oncology drugs live in medical claims — stack J-codes and NDCs.
- 3 · A cohort is a rule; publish the attrition table.
- 4 · LOT parameters go on the slide.
- 5 · Censoring is not death; a token is not anonymity.

Next week — Marketing Mix Modeling · Homework 1 released